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**A Data-Driven Optimization
Strategy to Decarbonize
Norwegian Oil & Gas Fields**

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MASTER THESIS

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Executive summary

In July 2024, global temperatures reached an unprecedented 17°C on two consecutive days (Copernicus Climate Change Service, 2024), underscoring the rapid pace of climate change. This warming is largely attributed to fossil fuel consumption, which accounts for over 75% of global greenhouse gas emissions (United Nations, 2024). Despite international climate efforts, oil demand remains steady at around 100 million barrels per day, with projections indicating only a slight decline by 2040 (International Energy Agency, 2022).

Norway, a key oil and gas producer supplying up to 3% of global demand, faces the dual challenge of reducing emissions while sustaining its energy output (Statistics Norway, 2024). The oil and gas sector accounts for 25% of Norway's domestic emissions, emphasizing the need for targeted strategies. Under the legally binding EU "Fit For 55" package, Norway is committed to reducing its emissions by 55% by 2030 (International Energy Agency, 2022).

This study evaluates two primary approaches: Electrification of offshore platforms and decommissioning of oil and gas fields using machine learning and optimization techniques. We conclude that 13 fields require electrification for Norway to be fit for 55. The other approach demands holding back production by 28% compared to emptying all current open fields. This strategic field-specific production reduction cuts Norwegian oil and gas lifetime emissions by 68%.

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1 Abstract

On July 21st and 22nd, 2024, the Earth recorded its two hottest days in recent history, with average temperatures reaching 17.15°C and 17.09°C, respectively (*Copernicus Climate Change Service*, 2024). These record-breaking temperatures highlight a pressing issue: our planet is warming rapidly along with the increasing atmospheric carbon dioxide concentration. The United Nations has identified fossil fuels—coal, oil, and gas—as the main contributors to this problem, responsible for over 75% of global greenhouse gas emissions and nearly 90% of all carbon dioxide emissions (*United Nations*, 2024). Despite this, global demand for oil remains steady at around 100 million barrels per day, with the International Energy Agency predicting a peak around 2030 in their Stated Policies Scenario, only to stabilize at 95 million barrels per day through 2040 (*International Energy Agency*, 2022).

Norway is a significant player in the global oil and gas market. While supplying between 2% and 3% of the world’s oil and gas, Norway has pledged to meet the EU’s emission reduction targets of the Paris Agreement and a 55% reduction by 2030 (*International Energy Agency*, 2022). With the oil and gas industry responsible for 25% of Norway’s domestic emissions (*Statistics Norway*, 2024), strategic efforts are essential and lacking for this industry to decarbonize. No phase-down strategy has been proposed by the Norwegian Government. In times of geopolitical uncertainty, there is an intensified focus on energy independence, hence it is of importance to find ways to minimize emissions while maintaining production.

By applying a data-driven approach with dual-objective linear programming optimization we aim to maximize production while minimizing emissions. Future emissions and production are estimated using Double Machine Learning. Using a Two-Way Fixed Effects model on 27 years of panel data from 10 electrified fields, our analysis reveals that electrification reduces emission intensity by approximately 40%, isolating this impact by controlling for both time-invariant and temporal factors. Furthermore, we highlight five different scenarios weighting emission reduction and production gains. Where the moderate production focus, with a production reduction of 7%, cuts emissions by 24%. The moderate emissions focus reduces production by 15%, and cuts emissions by 43%. In comparison, 3 and 13 fields would need electrification resulting respectively in emission reduction of 13% and 43%.

2 Introduction

The ongoing climate crisis, highlighted by record-breaking global temperatures in 2024, underscores the urgent need for effective strategies to reduce greenhouse gas emissions. In this context, the role of the oil and gas industry, particularly in major producing nations like Norway, becomes critical. As a significant contributor to global energy supply, Norway faces the challenge of balancing its role in the global energy market with its commitments to reducing emissions. The Norwegian oil and gas sector, responsible for a quarter of the nation's domestic emissions, stands at the forefront of this balancing interplay.

This thesis investigates the effectiveness of electrification and field decommissioning as strategies to reduce emissions from oil and gas operations on the Norwegian Continental Shelf (NCS). The aim is to provide actionable insights into optimizing emission reduction strategies while maintaining Norway's position as a key energy supplier. Given that the current remaining hydrocarbon reserves on the NCS are estimated to result in approximately 10 million tons of CO_2 equivalents (CO_2e) in emissions, a marginal improvement in field emission intensities are of significant importance.

The analysis presented in this study reveals that by limiting extraction to 85% of the remaining reserves from currently active fields, Norway could achieve a 43% reduction in total oil and gas-related emissions by 2050. Additionally, targeted electrification, could further reduce operational emission intensity by an average of 40%. These results highlight a pragmatic approach to decarbonizing the oil and gas sector, balancing environmental imperatives with the economic realities of continued energy production.

The research addresses the following key questions:

1. **What are driving the differences in emission intensities at Norwegian oil and gas fields?**
2. **What are the potential emissions savings from decommissioning, and how do they compare to those achievable through electrification?**

The results indicate that a combined approach of selective electrification and strategic decommissioning offers the most effective path forward. Electrification alone, particularly when targeted at high-emission fields, could achieve up to a 50% reduction in emissions with minimal impact on production. Yet, it would require significant power, either from offshore wind farms or through land-based grid connection. On the other side, decommissioning, while effective in reducing emissions, may have broader implications for global supply chains and market dynamics. This thesis provides critical insights for policymakers and industry stakeholders as they navigate the complexities of decarbonizing the oil and gas sector while maintaining economic viability.

3 Background Information

3.1 Importance of Oil and Gas in Norway

Norway is a significant player in the global oil and gas market. It is the largest producer of oil and natural gas in Western Europe, contributing 3% and 2.3% of the world's gas and oil consumption, respectively. The petroleum industry has been a cornerstone of Norway's economy since the discovery of oil in the North Sea in the late 1960s (*Norwegian Government*, 2021). The revenues generated from oil and gas have fueled economic growth and contributed to the country's welfare state through the Government Pension Fund Global, known as the Oil Fund. In 2021, the total export revenues from the petroleum industry were estimated to be over NOK 800 billion (*International Energy Agency. Energy Policy Review: Norway 2022*).

3.2 Current Emission Levels

The Norwegian oil and gas industry is a substantial source of greenhouse gas (GHG) emissions. In 2022, the sector was responsible for 25% (11.5 megatons of CO_2 equivalents) of the nation's domestic emissions *Statistics Norway: Emissions to air*. These emissions are primarily due to the reliance on gas turbines for power generation on offshore platforms. These turbines are used for drilling, pumping, and other operations. In addition a smaller proportion of emissions stem from venting, flaring and fugitives (*SP Global*, 2022).

3.3 The Case for Electrification

Electrification of offshore operations is seen as a promising strategy to reduce emissions. It is Equinor's primary decarbonization strategy (*Equinor*, 2024). By replacing gas turbines with electricity sourced from renewable energy such as hydro, wind, or solar power, the operational emission intensity can be significantly lowered, as seen in Figure 1. This approach involves supplying renewable energy through subsea cables to offshore platforms. As of August 2024 around eight fields have been fully or partly electrified, powered from onshore power and the offshore windfarm Hywind Tampen. These prospects have been showcasing reduced emission intensities compared to non-electrified fields (*Aker Solutions*, 2024).

3.4 The Norwegian Oil and Gas Industry

The Norwegian oil and gas industry operates on the NCS, which is divided into three main regions: the North Sea, the Norwegian Sea, and the Barents Sea. Each region has multiple oil and gas fields, characterized by different types of installations and infrastructure.

3.4.1 Installations

- **Fixed Platforms:** These are large, permanent structures anchored to the seabed, used for drilling, production, and processing.
- **Floating Production Systems:** These include Floating Production, Storage and Offloading units and semi-submersible platforms that can be moved and are typically used in deeper waters.
- **Subsea Systems:** These are installations located on the seabed, connected to fixed or floating platforms via pipelines and cables for processing and transport.

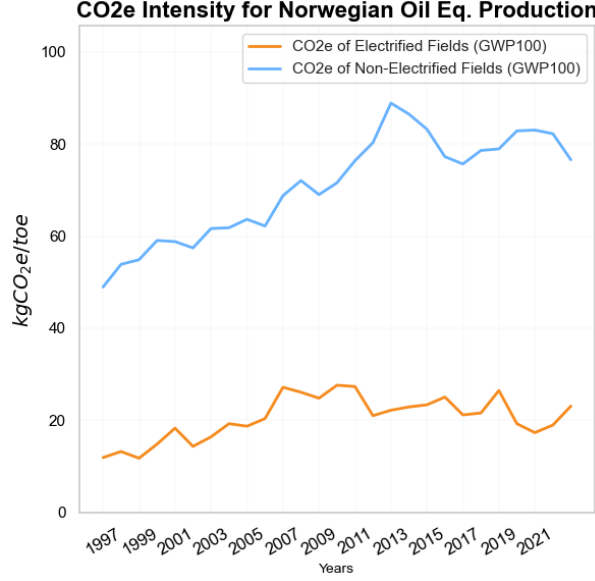


Figure 1: Emission Intensities for Electrified and Non-Electrified Fields on the NCS (Scope 1+2)

3.4.2 Operations and Infrastructure

Each oil field may have several rigs and multiple wells to maximize extraction efficiency. The fields are interconnected by an extensive network of pipelines, which transport oil and gas to onshore facilities for further processing and distribution. Some fields are purely subsea, where the extraction and initial processing occur on the seabed, and the hydrocarbons are transported to nearby fields or directly to shore through subsea pipelines (*Eivind Drabløs, Rystad Energy, 2024*).

3.5 Emissions Regulations and Policies

Norway has implemented stringent regulations to control emissions from the oil and gas sector, including the CO_2 Tax Act with a national carbon fee of 740 NOK and the Pollution Control Act limiting flaring to safety measures. These policies have led to Norway achieving one of the lowest emissions intensities for oil production globally (*Norwegian Petroleum, 2024*).

3.6 Global Warming Potential

This study examines both the Global Warming Potential over 100 years (GWP100) and 20 years (GWP20), incorporating CO_2 and methane (CH_4) emissions. This approach is essential for accurately assessing the climate impact of the oil and gas sector (*United States Environmental Protection Agency, 2024*).

3.6.1 Distributed Emissions and Production Share

In oil and gas operations, it is crucial to understand the concept of distributed emissions. Many oil fields do not have the facilities to process their own extracted hydrocarbons. Instead, they send the unprocessed oil (crude) and gas to a host processing facility. This arrangement means that the host facility bears the emissions burden for processing hydrocarbons extracted from multiple fields.

To accurately attribute emissions to their sources and understand the drivers of emission intensity differences across fields, it is necessary to calculate the production volume and emissions of field i ;

its associated processing facility H ; and all of H 's connected fields. With these calculations, we can allocate the emissions from the host processing facility back to the originating fields based on their share of the total production processed at the host facility.

$$\text{Distributed Emissions}_i = \frac{\text{Production}_i}{\sum_j \text{Production}_j} \times \sum_j \text{Emissions}_j$$

Where:

- $\text{Distributed Emissions}_i$ is the emissions attributed to field i .
- Production_i is the production of field i .
- $\sum_j \text{Production}_j$ is the total production of all fields j that use the same host facility H for processing.
- $\sum_j \text{Emissions}_j$ is the total emissions of all fields that use the same host facility H .

This formula ensures that the emissions are proportionally distributed based on the production share of each field. Whenever emission intensity is referenced, it should be understood as distributed emission intensity unless otherwise specified.

3.7 Local versus Global Measures

By replacing gas turbines with electricity sourced from renewable energy, operational emissions can be decreased by up to 40% (*Statistics Norway, 2024*). However, approximately 98% of the emissions from fossil fuels occur when they are burned by the end user, not during extraction and production. Even if Norway were to achieve net-zero emissions in its oil and gas operations, the burning of these fuels by end users around the world would still contribute substantially to global warming. If Norway were to extract all remaining oil and gas reserves from currently active fields—estimated at 1814 million tonnes of oil equivalent (toe)—and these were subsequently burned, it would add approximately 0.65 gigatonnes of CO_2 equivalent (CO_2e) to the atmosphere. This contrast between local emission reductions and global emission contributions necessitates a broader strategic approach. Figure 2 showcases the differences in emission intensities for electrified and non-electrified fields with Scope 3 emissions.

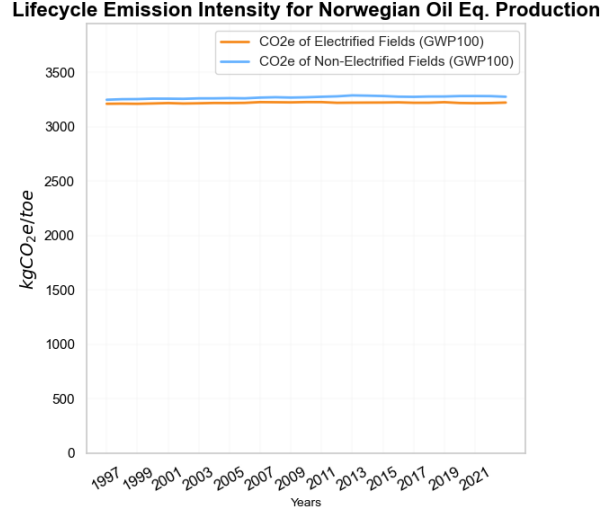


Figure 2: Lifecycle Emission Intensities for Electrified and Non-Electrified Fields on the NCS (Including Scope 3 emissions).

3.8 Strategic Importance and Future Prospects

Norway's commitment to the EU's emission reduction targets, as part of the 'Fit for 55' package, necessitates a strategic renewal of its oil and gas sector. The goal is to balance continued production with significant reductions in GHG emissions. By focusing on electrifying existing fields and optimizing the extraction process (*Energifakta Norge*, 2024). Norway lacks a clear strategy on its aims to achieve a 50% reduction in oil and gas-related emissions by 2030 and reach Net Zero in 2050. This transition is crucial for meeting international climate commitments and for ensuring the sustainability of the Norwegian oil and gas industry.

4 Related Literature

4.1 Emission Intensities and Fossil Fuel Demand

The field of energy research is vast, and its subcategory of energy intensity is broadly studied as well. The emission intensity of oil and gas as a topic belongs within this subsection (*Dixit, Y., El-Houjeiri, H., Monfort, J.C. et al., 2023*). While the historic oil and gas field-specific emissions and production volumes are well documented, a forward-looking optimization-based perspective has not been applied earlier. This study aims to fill in this literature gap and provide a pragmatic approach to align the Norwegian oil and gas industry with IEA’s energy and global warming scenarios. These three scenarios include:

- **Stated Policies Scenario (STEPS):** In the STEPS, global CO_2 emissions continue to rise slightly until peaking around 2040, leading to a modest reduction in emissions by 2050. The share of fossil fuels in primary energy demand decreases from 80% over the last two decades to 73% by 2030. Energy consumption increases by an annual average of 0.7% through to 2050. This scenario results in a projected temperature rise of around $2.7^{\circ}C$ by 2100. The global emission intensity under the STEPS is projected to increase by 2.2% annually from 2022 to 2030 (*International Energy Agency, 2024*).
- **Announced Pledges Scenario (APS):** The APS reflects the impact of all announced climate pledges. In the APS, global CO_2 emissions peak in the late 2020s and begin to decline gradually. By 2050, the share of fossil fuels in primary energy demand falls to 69%. Energy consumption peaks in the late 2020s and then starts to decline slowly. The APS leads to a temperature rise above $2^{\circ}C$ by 2100. The global emission intensity under the APS is projected to increase by 3% annually from 2022 to 2030 (*International Energy Agency, 2023*).
- **Net Zero Emissions by 2050 Scenario (NZE2050):** Achieving Net Zero emissions requires an almost 100% reduction in emissions from current levels. The share of fossil fuels in primary energy demand will drop to 62% by 2030. In this scenario, energy consumption declines by 1% per year from today, driven by significant improvements in energy efficiency and a major shift towards electrification, particularly in the mobility and heating sectors. This pathway is consistent with limiting global warming to $1.5^{\circ}C$. The global emission intensity under the NZE2050 is projected to increase by 4.1% annually from 2022 to 2030 (*International Energy Agency, 2023*).

4.2 Norwegian Oil and Gas Emission Intensity

Statistics Norway wrote a discussion paper in 2015 on CO_2 emissions from Norwegian oil gas extraction. The paper analyzed 74 individual fields over a times-pan of 16 years. The paper found that the average emission intensity of Norwegian oil and gas of 55kg CO_2 per ton oil equivalent (toe) is significantly lower than the global average of 130 kg CO_2 /toe (*Statistics Norway, 2015*).

The report identified several key factors influencing Norwegian emission intensities. Among these is the country’s robust carbon pricing strategy. Since the introduction of a CO_2 tax on the NCS in 1991, combined with the sector’s integration into the EU Emissions Trading System, companies operating in Norway have faced one of the highest carbon prices in the world. This economic pressure has driven significant reductions in emission intensities. Another critical factor highlighted in the report is the electrification of offshore platforms.

The Statistics Norway paper found that the production share of historic peak production (PSPP) played a significant role in determining emission intensity, with a 4.4 unit decrease in emission

intensity per unit of PSPP increase. The intuition is that as the production is high, the intensity is low. Additionally, in the early stages of a fields' lifetime, the subsea oil and gas pockets exhibits more pressure than towards its end. Hence a lower energy intensity and thus a lower emission intensity. Figure 3 depicts the differences in emission intensity across fields on the NCS.

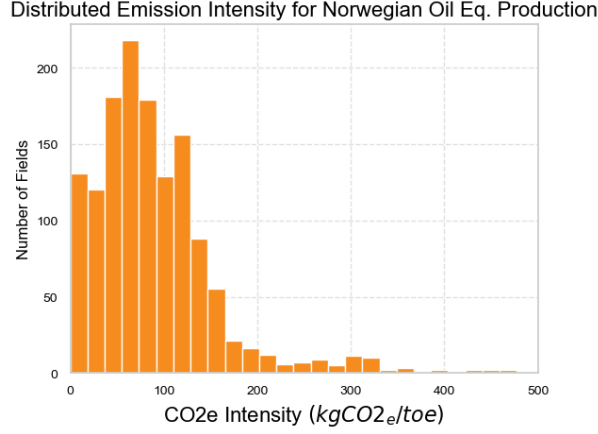


Figure 3: Distribution of Emission Intensities Showing Clear Differences

Furthermore, the gas reserve share shows the amount of the total original reserve is gas. Fields with a higher proportion of gas production typically showed lower emission profiles compared to those focused on oil. Given the estimate of 0.7 units decrease of emissions with one percentage change in gas reserve share, a field with purely oil exhibits twice the emission intensity as those with gas.

4.3 The Effect and Status of Electrification

In Norway, for example, there has been a strong push towards electrification driven by government policies and environmental regulations. The NCS has seen 8 projects implemented, and more are planned. A detailed assessment from (*Norwegian Ministry of Petroleum and Energy, 2021*) underscores that approximately 50% of emissions from Norwegian offshore activities could potentially be mitigated through electrification by 2030.

The study *An Integrated Assessment of the Environmental and Economic Impact of Offshore Oil Platform Electrification* by (*Riboldi et. al., 2019*) highlights that electrification of offshore platforms might reduce direct CO_2 emissions from oil and gas extraction processes. Specifically, the switch from gas turbines to electricity for power generation on platforms could lower emissions by up to 50% in certain scenarios. This reduction is primarily due to the decreased reliance on ineffective gas turbines and the increased efficiency of electrical power generation.

However, the study also points out that the environmental benefits depend significantly on the source of the electricity used. When the marginal effect of increased electricity demand is considered, the lifetime CO_2 emissions associated with the operation of the offshore facilities can increase by up to 40%, possibly due to the reliance on coal-fired power plants to meet this additional demand. Conversely, when the average effect is considered, emissions can decrease significantly, with potential reductions ranging from 48% to 90% depending on the geography and the energy mix.

Economically, while the initial investment for infrastructure and technology required for electrification is high, the long-term operational savings and carbon tax benefits present a plausible business case. The study notes that regions with stringent carbon pricing mechanisms stand to gain the most economically from electrification due to avoided penalties and enhanced operational efficiency.

On the other side, the Statistics Norway paper points out that upfront capital expenditure of electrification exceeds the reduction in carbon taxes. These calculations are heavily dependant on the carbon price and the lifetime of the field.

4.4 Why Norway is Eligible for Electrification

Norway is particularly well-suited for the electrification of offshore oil platforms due to its energy mix, which is dominated by renewable energy, especially hydropower. Norway generates 92% of its electricity from hydropower, making it one of the cleanest energy producers in Europe. This high percentage of renewable energy means that the additional electricity required for offshore platform operations can be supplied with minimal additional CO_2 emissions. This is in stark contrast to countries that rely heavily on fossil fuels for electricity generation, where the marginal increase in power demand could result in significant emissions increases (*International Energy Agency*, 2022).

Furthermore, Norway's extensive hydro power infrastructure, with a generation capacity of around 33.7 GW, and huge storage capabilities, which account for half of Europe's reservoir capacity, provide a reliable and sustainable energy source for electrification projects. This makes Norway a suitable model for the benefits of electrification in reducing operational oil and gas emissions. It is important to add that there are concerns about the additional demand for electricity will inflate Norwegian electricity prices (*Norwegian Parliament*, 2022). If Norway's electricity supply exceeds its demand, the country might be forced to import less sustainable energy sources, such as coal from Germany. In such a scenario, the emission intensity and absolute operational emissions will drastically increase.

Another ongoing discussion in the context of Norway's energy strategy circles around the opportunity costs of using renewable electricity for offshore electrification, particularly when this energy could be allocated to other critical activities, such as hydrogen production. The potential for producing green hydrogen (hydrogen produced via electrolysis using renewable electricity) in Norway is significant. If the electricity currently directed toward offshore electrification were instead used to produce hydrogen, Norway could substantially ramp up its hydrogen production capabilities. This would not only support domestic decarbonization efforts but also position Norway as a major exporter of green hydrogen, contributing to global emission reduction targets.

5 Assumptions

This section outlines the key assumptions underlying the research methodology and analysis within this thesis. These assumptions are essential for interpreting the findings and understanding the scope and limitations of the study.

5.1 Assumptions Related to Electrification

- **Renewable Energy Supply:** The analysis assumes that the energy replacing gas turbine-generated power during electrification is sourced entirely from renewable energy. This assumption leverages Norway’s extensive renewable energy infrastructure, predominantly hydroelectric power, ensuring that electrification does not introduce additional GHG emissions.
- **Estimation of Electrification Effects:** Electrification’s impact on emissions is quantified using a Two-Way Fixed Effects (TWFE) model. This model compares the average emissions of non-electrified fields with the reductions attributed to electrification, as captured by the model’s coefficients. It assumes that these reductions are solely due to electrification, with other variables held constant, and captured by the TWFE model.
- **Partial Electrification Feasibility:** Given the significant power demands of full electrification, the study assumes a partial electrification scenario. This reflects current industry realities, where fully decarbonizing oil and gas operations is challenging. The partial electrification assumption is therefore considered more practical and representative of likely outcomes.
- **Alternative Uses of Electricity:** The analysis does not account for alternative uses of the electricity, such as its application in producing green hydrogen. Instead, it focuses exclusively on emission reductions from the electrification of oil and gas fields, without comparing potential benefits across different sectors.
- **Long-Term Electrification Commitment:** It is assumed that once a field is electrified, it remains so for the remainder of its operational life. The study does not consider scenarios where electrification might be reversed or altered, thereby assuming a consistent application of renewable energy throughout the field’s lifespan.

5.2 Assumptions Related to Field Decommissioning and Emission Reductions

- **Scope of Emission Intensity Optimization:** The optimization of emission intensity considers only Scope 1 and Scope 2 emissions associated with exploration and production (E&P) activities. Scope 1 emissions cover direct emissions from field operations, while Scope 2 covers indirect emissions from purchased energy. Scope 3 emissions, which include all other indirect emissions across the value chain, are not included in this analysis. Emission reduction percentages discussed refer exclusively to these operational emissions unless otherwise specified.
- **Business-As-Usual Benchmark:** The baseline scenario assumes that all currently operational fields continue production without any decommissioning. This business-as-usual scenario serves as the benchmark for evaluating the impact of field decommissioning on emissions, and all emission reduction numbers stems from comparing to this benchmark.
- **Focus on Existing Fields:** We focus exclusively on fields that are currently operational. Although Norway has issued permits for 19 new oil and gas fields, these future developments does not have enough data to be included in this analysis (*Norwegian Government*, 2023). Consequently, the study does not account for potential future emissions and production from these upcoming fields.

- **Global Market and Supply Implications:** The study assumes that a reduction in Norway’s oil and gas production due to decommissioning does not account for potential global market adjustments. Specifically, it does not consider how other countries might increase production to meet global demand. Given Norway’s relatively low emission intensity, this could result in higher global emissions if less efficient producers fill the gap.

5.3 Additional Assumptions

- **Data Integrity and Modeling Assumptions:** The analysis assumes that the data used, including production volumes, emission records, and energy consumption metrics, are accurate and reliable. Furthermore, the models employed are assumed to be appropriate for capturing the relationships between electrification, decommissioning, and emissions, with any limitations adequately controlled. We have used standard metrics of conversion supported by the The Intergovernmental Panel on Climate Change, International Energy Agency, and Norwegian Petroleum.
- **Stability of Technological and Policy Contexts:** The study assumes no significant technological advancements or policy changes that could alter the feasibility or impact of electrification and decommissioning during the study period. This includes stable energy prices, the continued availability of renewable resources, and a consistent regulatory environment governing Norway’s oil and gas sector.

6 Methods and Analysis

This thesis integrates quantitative analysis, qualitative insights, and literature review to examine carbon intensity and emission reduction through electrification in the Norwegian oil and gas sector. The integration of these methods allows for a nuanced exploration of the research questions². The methods section explores the qualitative, but emphasises the quantitative aspects of this thesis.

6.0.1 Quantitative Analysis

Advanced econometric and machine learning techniques form the core of the quantitative analysis. Double Machine Learning (DML) is used to estimate the causal effects of field-specific parameters. Additionally, a Two-Way Fixed Effects model isolates the impact of electrification on emissions, controlling for field-specific and temporal factors. These models provide critical empirical insights into the factors driving emissions and inform projections of future scenarios.

6.0.2 Qualitative Insights

Expert interviews with industry leaders, including Eivind Drabløs from Rystad Energy, William Brekke from Pareto Securities, and Sigrid da Costa from Harbour Energy, provide qualitative data that contextualize the quantitative outcomes. These interviews offer practical perspectives on operational strategies, investor sentiment, and technological advancements, enriching the overall analysis. The interview with Eivind Drabløs was particularly influential in guiding the selection of variables for the econometric TWFE model.

6.0.3 Literature Review

The literature review identified gaps⁴ in current knowledge and situates the thesis within broader academic and industry discussions.

6.1 Data Sources

This chapter outlines the data sources utilized in this research, the methods employed for data processing, and the techniques applied for analysis and optimization. The focus is on how these data sources inform the assessment of the impact of electrification on emissions from oil and gas operations. The chapter covers the origins and characteristics of the datasets, the preprocessing steps, and the modeling and optimization techniques. The entire dataset consists of 100 columns capturing years from 1997 to 2022. The entire code and data can be accessed through GitHub¹.

The code is structured in five different notebooks:

1. **Data Building** Includes data gathering and merging.
2. **Data Cleaning** Includes removal, renaming and conversion of various variables.
3. **Data Processing** Includes calculations of distributed emissions, emission intensities, GWP100 and GWP20.
4. **Data Modeling** Includes production and emission calculations until 2050 or until the fields reserve is empty. These are derived from regression- and double machine learning models.
5. **Data Optimization** Includes linear programming and optimization techniques to modeling future production and emissions.

¹https://github.com/percw/Norwegian_oil_gas_decarbonization

6.1.1 Production Data

The production data utilized in this research is sourced from authoritative Norwegian databases, primarily The Norwegian Offshore Directorate² and Norwegian Petroleum³. These databases provide comprehensive records of oil and gas production volumes on monthly basis. This granularity is used to compute the intra-year standard deviation. To match the annual emissions data structure, yearly emissions are calculated.

Key production metrics include:

- **Net Oil Equivalent Production** This metric consolidates different types of hydrocarbons, such as crude oil, natural gas, and condensates, into a standardized unit of millions of standard cubic meters (Sm^3) per year.
- **Original Reserve Size** The original reserve size of a field is used to assess the field's potential for production and its long-term viability. This metric represents the estimated total quantity of recoverable hydrocarbons present in the field at the time of its discovery.
- **Current Remaining Recoverable Oil Equivalent:** This metric provides an estimate of the remaining recoverable reserves within each field. It is a critical indicator for assessing the sustainability of production and for planning long-term strategies, including the potential for electrification to extend field life while reducing emissions.
- **Historical Production Trends:** The dataset includes detailed historical production data, allowing for the analysis of production efficiency over time and the assessment of how past operational changes.

6.1.2 Emissions Data

The emissions data, encompassing both CO_2 and CH_4 emissions, is sourced from The Norwegian Environment Agency⁴. These emissions are reported by field operators and verified by the agency to ensure accuracy. This data is crucial for quantifying the environmental impact of oil and gas operations and for assessing the effectiveness of various emission reduction strategies.

Key emission metrics include:

- **CO_2 Emissions (Tonnes):** This metric tracks the total annual CO_2 reported emissions from each field, measured in tonnes.
- **Methane (CH_4) Emissions:** Methane emissions, a significant component of GHG emissions due to CH_4 's high global warming potential. The dataset includes both fugitive emissions (unintentional leaks) and operational emissions (venting and flaring). The dataset does not distinguish between these emission sources.
- **Emission Intensity ($kgCO_{2e}/toe$):** The emission intensity of production, a key indicator of environmental performance, is calculated as follows:

$$CO_2 \text{ Emission Intensity (kg } CO_{2e}/toe) = \frac{\text{Total } CO_{2e} \text{ Emissions (kg)}}{\text{Ton Oil Equivalent (toe)}}$$

This intensity metric allows for a normalized comparison of emissions across fields and over time, providing insights into the efficiency of emission reduction measures.

²<https://www.norskeutslipp.no/en/Frontpage/>

³<https://www.norskipetroleum.no/en/>

⁴www.norskeutslipp.no

6.1.3 Field Data

The field data provides detailed insights into the physical and operational characteristics of oil and gas fields, including wells, facilities, infrastructure, and investments.

- **Wells:** The dataset includes information on the number, type, depth and status of wells in each field, including production wells, injection wells, and exploration wells. The operational status of these wells (e.g., producing, plugged, abandoned) is also recorded, providing a detailed overview of field activity.
- **Facilities:** Detailed descriptions of the production and processing facilities within each field are provided, including . This information is vital for assessing the infrastructure’s capability to integrate electrification technologies.
- **Field Classification:** Fields are categorized by type (e.g., Multi-Well, FPSO, FSU, Spar and Jacket) and operational status (eg., shut down and in service), as well as water depth and lifetime.
- **Investments:** Data on historic and projected capital expenditures highlight the financial resources allocated to developing and maintaining these fields.
- **Pipelines and Infrastructure:** The dataset includes information on the pipeline networks and other infrastructure that connect fields to processing facilities and export terminals. Understanding this infrastructure is crucial for evaluating the distributed emissions.

6.2 Causal Discovery

DirectLiNGAM has been employed to estimate the causal relationships between key operational and environmental variables related to oil and gas production. This method allows for a deeper understanding of how production, field characteristics, and technological investments influence emission outcomes.

DirectLiNGAM is well-suited for uncovering causal structures in systems where relationships among variables are linear but may involve non-Gaussian noise. This method builds upon the principles of Independent Component Analysis (ICA) to separate mixed signals into their independent components, making it possible to identify causal directions in complex datasets as demonstrated in ALCM: Autonomous LLM-Augmented Causal Discovery Framework (*Elahe Khatibi et al., 2024*). By applying a threshold to the relationships identified by DirectLiNGAM, one can filter out weaker associations and focus on the most significant causal pathways, as illustrated in the graph presented.

This machine learning method has been crucial to understand the complex dataset and provided a clear direction from the beginning. The LLM-driven causal discovery highlighted the following connections for distributed emission intensity and production share emissions:

- **Distributed Emission Intensity:**
 - Original recoverable oil
 - Original recoverable gas
- **Production Share Emissions:**
 - Current remaining oil eq.
 - CO_2e emissions
 - Share of original reserve

– Oil eq. production

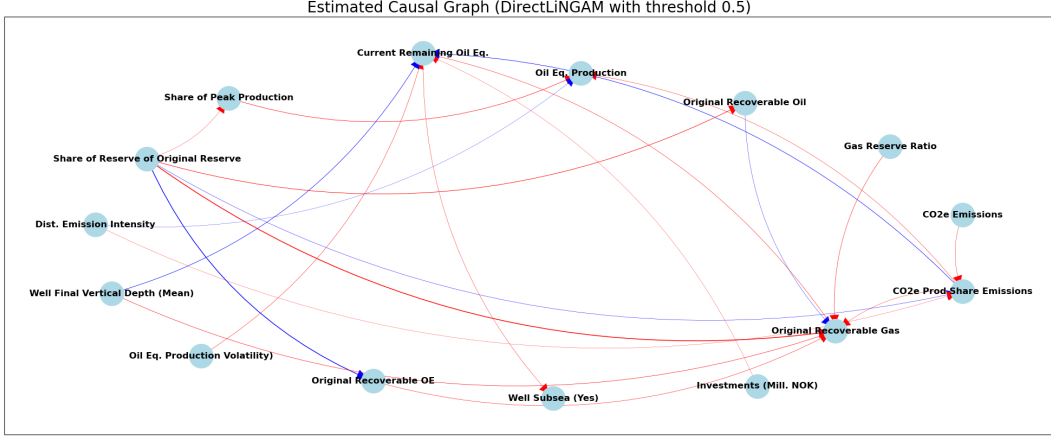


Figure 4: Causal Discovery using DirectLiNGAM

6.3 Double Machine Learning (DML)

6.3.1 Introduction to DML

Double Machine Learning (DML) is a robust statistical technique due to its effectiveness to estimate causal effects in the presence of high-dimensional data. The method gained traction after the paper *Double/Debiased Machine Learning for Treatment and Causal Parameter* was published (Victor Chernozhukov et al., 2016). DML has since been more widely available to the public through guides such as *Causal Inference for the Brave and True* by Matheus Facure Alves [2] and *Double Machine Learning Simplified* by Jacob Pieniazek [14]. DML handles the potential biases introduced by confounding variables by combining machine learning models with traditional econometric methods. The data for this analysis consists of detailed field-level reported information from the Norwegian oil and gas sector, including measurements of facility types, rig types, and wells, future and historic investments. In this study, DML is utilized to estimate the causal relationships between emission intensity and three key field-specific parameters: Remaining Reserve, Share of Peak Production, and Total Production Volume.

6.3.2 DML Specification

The DML approach involves two main stages:

1. **Nuisance Parameter Estimation:** Gradient Boosting Machines (GBMs) were used to estimate the nuisance parameters due to their ability to capture complex, non-linear relationships. These models predict the outcome (emission intensity) and the treatment variables (remaining reserve, share of peak production, total production volume) as functions of the covariates. The configuration for the GBMs included a maximum depth of 3, 100 estimators, and a learning rate of 0.1.
2. **Orthogonalization and Estimation:** The residuals from the first stage were used to form orthogonalized versions of the treatment and outcome variables. A second-stage regression, using Ordinary Least Squares (OLS), was then performed to estimate the causal effect of the treatment variables on the outcome. This regression included the quadratic terms to capture the non-linear relationships identified.

6.3.3 Fitted Quadratic Equations

To capture the non-linear relationships between emission intensity and the field-specific parameters, fitted quadratic equations were employed for each of the three parameters: Remaining Reserve, Share of Peak Production, and Total Production Volume. The quadratic terms allow the model to account for potential curvilinear effects, providing a more accurate representation of the underlying relationships.

The fitted quadratic equations used in the DML framework are as follows:

$$\text{Emission Intensity}_i = \beta_0 + \beta_1 \cdot X_{1i} + \beta_2 \cdot X_{1i}^2 + \epsilon_i \quad (1)$$

$$\text{Emission Intensity}_i = \beta_0 + \beta_3 \cdot X_{2i} + \beta_4 \cdot X_{2i}^2 + \epsilon_i \quad (2)$$

$$\text{Emission Intensity}_i = \beta_0 + \beta_5 \cdot X_{3i} + \beta_6 \cdot X_{3i}^2 + \epsilon_i \quad (3)$$

Where:

- X_{1i} represents the Remaining Reserve for field i .
- X_{2i} represents the Share of Peak Production for field i .
- X_{3i} represents the Total Production Volume for field i .
- $\text{Emission Intensity}_i$ is the emission intensity for field i .
- ϵ_i is the error term.

6.4 Two-Way Fixed Effects Model (TWFE)

6.4.1 Introduction to TWFE

TWFE is applied to estimate the impact of electrification on emission intensity. Unlike DML, which is highly effective in high-dimensional settings, TWFE explicitly accounts for entity-specific and time-specific fixed effects. This makes TWFE more appropriate for panel data where the focus is on isolating the impact of a treatment (electrification) while controlling for unobserved, time-invariant characteristics of each field and temporal shocks that might affect all fields simultaneously.

6.4.2 TWFE Specification

The PanelOLS function from the *linearmodels*⁵ library was used to fit the TWFE models, with clustered standard errors to account for within-field correlation. The TWFE model was implemented using the following steps:

1. Dependent Variable:

- Distributed Emission Intensity

2. Control Variables:

- Net Oil Equivalent Production Yearly (millions of standard cubic meters)
- Share Reserve of Original Reserve

⁵<https://pypi.org/project/linearmodels/>

The variables *Net Oil Production Yearly* and *Net Gas Production Yearly* were initially considered but removed due to increased p-values, indicating they were capturing the electrification effect. *Share of Peak Production* and *Original Recoverable NGL* were also removed as they were fully absorbed by the model.

3. **Data Preparation:** The dataset was filtered to include only the necessary columns for the analysis: field ID, year, electrified status, dependent variables, and control variables. The 'year' column was converted to an integer format, and any rows with missing values in the dependent or control variables were dropped.
4. **Creating Year Dummies:** To account for time-fixed effects, year dummies were created and added to the dataset.
5. **Panel Data Setup:** The dataset was set up as a panel data structure with a multi-index of field ID and year to facilitate the panel data analysis.
6. **Model Estimation:** The two TWFE models were estimated using the following formula with and without controls:

$$\text{Emission Intensity} \sim \text{electrified} + \text{year_dummies} + \text{EntityEffects} \quad (4)$$

$$\text{Emission Intensity} \sim \text{electrified} + \text{control_variables} + \text{year_dummies} + \text{EntityEffects} \quad (5)$$

6.5 Future Production and Emissions

This section outlines the methodology used to project future production volumes and associated emissions for oil fields on the Norwegian Continental Shelf, integrating parameters derived from the DML and TWFE models.

6.6 Production Forecasting

The future production for each field is projected from 2024 to 2050 using a model that incorporates dynamic adjustments based on the field's remaining reserves. The model integrates output from the DML and TWFE models to refine its estimates.

1. **Ramp-Down Threshold:** A ramp-down threshold is established at 10% of the original recoverable oil. When the remaining reserves fall below this threshold, production begins to decline proportionally.
2. **Yearly Production Calculation:** For each year i , the production P_i is calculated as follows:

$$P_i = \begin{cases} \text{median_production}, & \text{if } R_i \geq \text{median_production} \\ R_i, & \text{if } R_i < \text{median_production} \end{cases}$$

Where R_i represents the remaining reserves at the start of year i . The cumulative production C_i by the end of year i is updated as:

$$C_i = C_{i-1} + P_i$$

The process iterates until $R_i = 0$, signaling the depletion of the field's reserves.

6.7 Emission Intensity Adjustment

The future emissions for each field are calculated by dynamically adjusting the baseline emission intensity using parameters derived from the DML and TWFE models.

1. **Baseline Emission Intensity:** The baseline emission intensity E_{baseline} is the mean historical emission intensity for each field.
2. **Dynamic Adjustments:** For each year i , the adjusted emission intensity E_i is calculated as:

$$E_i = E_{\text{baseline}} \times \left(1 + \frac{a \times R_i + b \times \frac{P_i}{P_{\text{peak}}} + c \times P_i}{3} \right)$$

Where:

- R_i is the remaining reserve percentage.
- $\frac{P_i}{P_{\text{peak}}}$ is the share of peak production.
- P_i is the production volume for year i .
- a, b, c are coefficients derived from the DML and TWFE models.

3. **Total Emissions Calculation:** The total emissions T_i for each year i are calculated as:

$$T_i = P_i \times E_i$$

This equation accounts for the production in year i and the dynamically adjusted emission intensity.

6.8 Production and Emissions Optimization

The goal of this optimization is to balance the trade-off between maximizing production and minimizing emissions from fields on the NCS. We introduce a parameter λ that allows us to control the weight given to production versus emissions in the objective function. The optimization can be described in the following steps:

1. **Normalization:** Normalize the production and emissions data to ensure they are on comparable scales. Let P_i be the total lifetime production for field i and E_i be the total lifetime emissions for field i . Normalize these values to obtain P'_i and E'_i :

$$P'_i = \frac{P_i - P_{\min}}{P_{\max} - P_{\min}}$$

$$E'_i = \frac{E_i - E_{\min}}{E_{\max} - E_{\min}}$$

2. **Decision Variables:** Define binary decision variables x_i for each field i , where $x_i = 1$ if the field is selected, and $x_i = 0$ otherwise.
3. **Objective Function:** Let P_i be the total lifetime production for field i , given by the sum of annual production over the field's lifetime:

$$P_i = \sum_{t=1}^{T_i} p_{i,t}$$

Let E_i be the total lifetime emissions for field i , given by the sum of annual emissions over the field's lifetime:

$$E_i = \sum_{t=1}^{T_i} e_{i,t}$$

Construct the objective function to maximize a weighted sum of normalized production and normalized emission reduction. The parameter λ (where $0 \leq \lambda \leq 1$) controls the trade-off.

The objective function Z can be written as:

$$Z = \sum_{i=1}^n x_i (\lambda P'_i - (1 - \lambda) E'_i)$$

Here, λ determines the relative importance of production versus emissions. When $\lambda = 1$, the focus is entirely on production, where all fields will be kept open. When $\lambda = 0$, the focus is entirely on minimizing emissions by shutting down all fields.

4. **Constraints:** - Ensure that the decision variables x_i are binary:

$$x_i \in \{0, 1\} \quad \forall i$$

By systematically adjusting λ , we can analyze the trade-offs between maximizing production and minimizing emissions, providing valuable insights for decision-making in the management of oil fields.

6.9 Electrification Optimization

The optimization process is designed to determine the optimal electrification strategy across various fields, considering different levels of electrification and weighting factors (lambdas) that balance production and emissions.

1. **Objective Function:** The objective of the optimization is twofold:
 - **Maximize Total Production:** Ensuring that production remains at its highest potential.
 - **Minimize Total Emissions:** Reducing the carbon footprint by considering electrified and non-electrified scenarios.

The objective function is formulated as follows:

$$\text{Maximize } Z = \sum_{i=1}^n (\lambda \times P_i - (1 - \lambda) \times E_{\text{elec},i} + (1 - \lambda) \times E_{\text{non-elec},i})$$

Where:

- Z is the objective function value.
- λ is the weighting factor for production versus emissions.
- P_i is the total lifetime production for field i .
- $E_{\text{elec},i}$ is the total lifetime emissions if field i is electrified.
- $E_{\text{non-elec},i}$ is the total lifetime emissions if field i is not electrified.

2. **Electrification Levels:** Different electrification levels are tested, representing the percentage of fields to be electrified (e.g., 0%, 5%, 10%, 15%, 20%). For each level, the number of fields to electrify is calculated as:

$$\text{Number of Electrified Fields} = \text{Electrification Level} \times \frac{\text{Total Fields}}{100}$$

3. **Decision Variables:** Binary decision variables e_i are introduced for each field i , where:

$$e_i = \begin{cases} 1, & \text{if field } i \text{ is electrified} \\ 0, & \text{if field } i \text{ is not electrified} \end{cases}$$

4. **Constraints:** A key constraint ensures that the number of electrified fields matches the specified electrification level:

$$\sum_{i=1}^n e_i = \text{Number of Electrified Fields}$$

5. **Solver Implementation:** The optimization problem is solved using the OR-Tools solver, specifically the "SCIP" solver, which is well-suited for handling integer programming problems. The solver iterates over the possible values of lambda and electrification levels to identify the optimal electrification strategy.
6. **Solution Evaluation:** The solver assesses the optimality of each solution, categorizing them as either optimal, feasible, or infeasible. The solutions include:
- **Optimized Emissions:** Calculated by applying the electrification decision to each field's emissions.
 - **Optimized Production:** Remains constant as electrification does not affect production directly.

6.10 Interviews

This chapter presents the output from three expert interviews conducted to gain deeper insights into various aspects of the Norwegian oil and gas sector, including production dynamics, market sentiment, and exploration activities. The interviews were conducted with key industry stakeholders: Eivind Drabløs, Vice President and Product Manager at Rystad Energy; William Brekke, Partner in Investment Banking at Pareto Securities; and Sigrid da Costa, Explorationist at Harbour Energy. Their perspectives provide valuable qualitative data that complement the quantitative analyses and literature review in this thesis.

6.10.1 Norwegian Oil and Gas Production Dynamics

Eivind Drabløs, an industry expert from Rystad Energy, provided detailed insights into the fixed and dynamic features of oil fields and rigs on the NCS. According to Drabløs, several variables can change over the life of an oil field, including the oil-to-gas ratio, the number of rigs, and the original reserve estimates as new wells are drilled. These factors contribute to fluctuations in production and can be managed to optimize output depending on external conditions such as market demand and maintenance schedules.

Drabløs confirmed that production can be adjusted within a year, depending on operational needs or economic conditions, although it is more common to produce at maximum capacity to ensure economic efficiency. He also noted that while fields usually don't hold off production for extended periods, it can occur in exceptional cases involving significant field repairs or upgrades.

One of the most critical factors influencing carbon intensity, according to Drabløs, is the age and remaining reserves of the field. Older fields, with lower remaining reserves, tend to have higher carbon intensities due to the increased energy required to extract oil and gas as pressure declines. He highlighted that electrification would, in his experience have the biggest influence on field emissions.

6.11 Financial Services and Market Sentiment

William Brekke of Pareto Securities provided an insightful overview of the current investment sentiment towards the global energy sector, distinguishing between renewable and fossil fuel energy sources. Brekke highlighted that while the demand for fossil fuels remains robust due to stable global energy needs and recent supply-side disruptions, there is an increasing reluctance among investors to finance traditional fossil fuel projects. This reluctance is largely driven by growing environmental, social, and governance (ESG) pressures, which are becoming a critical factor in the decision-making process for both investors and companies.

This shift in investment focus is particularly relevant to the Norwegian oil and gas industry, where efforts to reduce carbon intensity through electrification and other means are crucial for maintaining investor confidence. As Brekke noted, the decline in debt funding for oil and gas projects in Europe—despite their potentially high returns—signals a broader industry trend towards sustainability. Investors are increasingly inclined to allocate capital to clean-tech bonds and sustainable energy projects, aligning their portfolios with the global push towards lower carbon emissions.

Brekke's insights underscore the importance of electrification and other carbon-reduction strategies in securing future investments in the Norwegian oil and gas sector. By reducing the carbon intensity of oil production through electrification, Norwegian Exploration and Production (E&P) companies can position themselves as attractive options for ESG-conscious investors. The relatively low carbon intensity of Norwegian oil and gas operations, compared to other jurisdictions, is a significant competitive advantage in a market where regulatory stability and environmental performance are becoming as important as financial returns.

Moreover, Brekke emphasized the stability and predictability of Norway's regulatory environment, which has been instrumental in maintaining investor confidence. The Norwegian petroleum tax system, with its consistent marginal tax rate of 78% since 1992, combined with favorable cost recovery provisions, supports long-term investments in the sector. The flexibility demonstrated by the authorities, particularly during the 2020 oil price shock, further highlights Norway's commitment to sustaining its oil and gas industry while navigating the transition to cleaner energy.

These financial and regulatory factors are critical for understanding how Norway's oil and gas sector can continue to attract investment while pursuing electrification and other carbon-reduction strategies. By aligning with investor priorities—such as low carbon intensity and regulatory stability—Norwegian E&P companies can secure the funding necessary to implement the technological innovations that will reduce their environmental impact.

6.11.1 Exploration and Seismic Analysis

Sigrid da Costa, an explorationist at Harbour Energy, provided insights into the evolving demand for seismic subsea data analysis on the NCS. Da Costa explained that the demand for high-quality

seismic data has increased as companies seek to extract value from more complex and challenging geological structures, particularly in less explored areas such as the Barents Sea and the Norwegian Sea. She noted that while traditional seismic data is still valuable, there has been a significant shift towards more advanced technologies and methodologies, including machine learning and new software platforms, to enhance the accuracy and efficiency of seismic interpretation.

Da Costa highlighted that the focus of exploration has shifted from well-known areas in the North Sea to newer regions where fewer wells have been drilled historically. This shift is driven by the need to sustain production levels as more mature fields decline. The potential development of infrastructure, such as pipelines to the Barents Sea, is expected to further boost exploration activities in these regions.

The interviews with Drabløs, Brekke, and da Costa provide a comprehensive view of the current state and future prospects of the Norwegian oil and gas sector. Their insights underline the dynamic nature of production management, the shifting investment landscape towards sustainability, and the technological advancements in exploration, all of which are critical for understanding the sector's ongoing evolution and its role in the global energy transition.

7 Results

7.1 DML to Estimate Lifetime Field Emission Intensity

7.1.1 DML 1 - Share of Remaining Reserve on Emission Intensity

The DML analysis was conducted to estimate the relationship between the Remaining Reserve Percentage and Emission Intensity. The results highlight the impact of this variable on both emission intensity and total emissions across the fields studied during the fields lifetime.

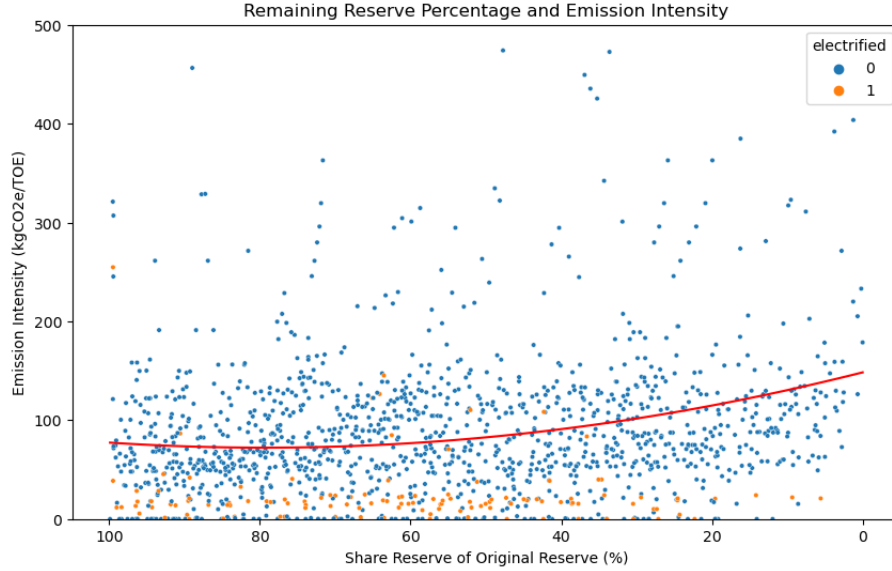


Figure 5: Remaining Reserve and Distributed Emission Intensity

Key Findings The coefficient for *emission intensity distributed* is -0.45, with a standard error of 0.11. This result is statistically significant. The negative coefficient suggests that the initial phase of the field where the remaining reserve is high yields lower emission intensities. Interestingly, the total emissions coefficient is positive, although not statistically significant it does imply that the start-up phase is energy consuming.

The quadratic relationship for emission intensity is described by the equation:

$$y = 0.01x^2 - 1.92x + 148.36$$

This equation indicates a non-linear relationship where emission intensity initially slightly decreases as the remaining reserves diminish, but this effect tapers off as reserves continue to decline. As Eivind Drabløs (Vice President at Rystad Energy) pointed out in our interview, the production is usually higher in the start, as the field exhibits a higher natural pressure; both boosting the production and reducing the required energy to create suction and pull. As the field matures, water is used to create pressure to extract the remaining oil and gas. This process is energy intensive while the production diminishes, hence the expected emission intensity rises.

Summary Table

The results from this analysis provide a solid understanding of how remaining reserves influence the emission intensity. The negative relationship between remaining reserves and emission intensity suggests that as oil fields deplete, their emission intensities increases. However, the relationship between remaining reserve and total emissions is positive. These discoveries are crucial for developing

	Emission Intensity	Emissions
Coefficient	-0.45	197.39
Standard Error	0.11	190.95
T-Statistic	-4.02	1.03
P-Value	5.00×10^{-5}	0.3

Table 1: Summary of DML Results for Remaining Reserve Percentage and Emission Intensity

strategies that optimize production while minimizing emissions, particularly as fields approach the end of their lifecycle.

7.1.2 DML 2 - Share of Peak Production on Emission Intensity

The DML analysis was conducted to estimate the relationship between the Share of Peak Production and Emission Intensity. The results highlight the different impact of this variable on both emission intensity and total emissions across the fields studied during their lifecycle.

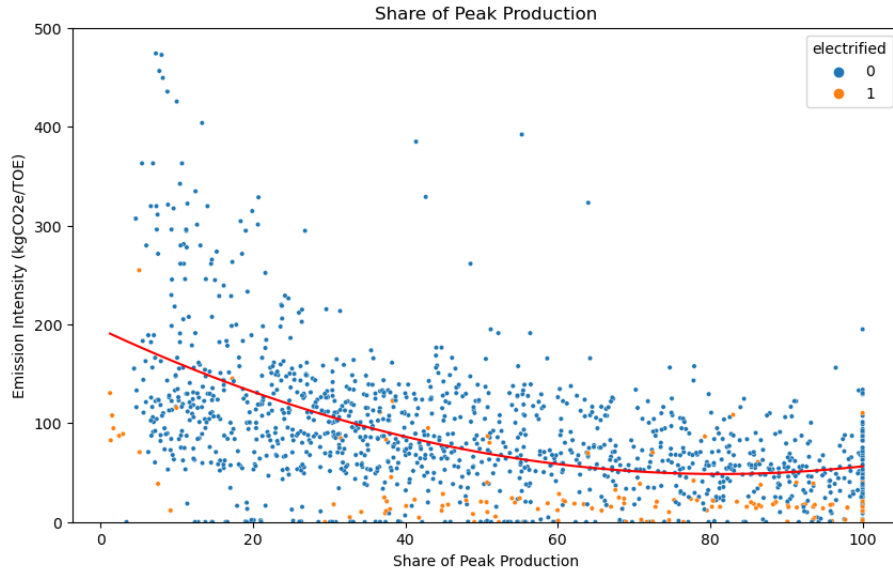


Figure 6: Share of Peak Production and Emission Intensity

Key Findings The coefficient for *emission intensity distributed* is -0.61, with a standard error of 0.11. This result is statistically significant, with a P-value near zero. The negative coefficient suggests that an increase in the share of peak production is associated with a decrease in emission intensity. Also statistically significant is the total *emissions*, which in contrast with the intensity has a positive coefficient. Higher production yields lower intensities, but on the cost of total emissions.

The quadratic relationship for emission intensity is described by the equation:

$$y = 0.02x^2 - 3.60x + 195.00$$

This equation indicates a non-linear relationship where emission intensity decreases as production approaches its peak but then levels off as production increases further. The constant at 195 indicates that as the production goes to zero, the emission intensity goes to 195 kgCO₂e/toe, which is far above the global average emission intensity (*Statistics Norway*, 2015). Furthermore compared to

reserve share, it is interesting to see that the effect on emission intensity is stronger for this intra-field parameter that measures to what extent of the maximum capacity the fields are able to produce at. For policymakers and operators it translates to putting in place technology and legislation that enables high production until the field approaches decommissioning.

Summary Table

	Emission Intensity	Emissions
Coefficient	-0.61	1566.06
Standard Error	0.11	176.31
T-Statistic	-5.40	8.88
P-Value	6.50×10^{-8}	6.50×10^{-19}

Table 2: Summary of DML Results for Share of Peak Production and Emission Intensity

The results indicates that when production levels drop to approximately 27% of their peak, the emission intensity doubles (113 kg CO_2 /toe) compared to the intensity observed at peak production (56 kg CO_2e /toe). This sharp increase in emission intensity as production declines illustrates the inefficiencies that arise when a field operates significantly below its maximum capacity.

The DML analysis was also applied to examine the relationship between oil equivalent (oe) production—comprising condensate, NGL, crude, and gas—and emission intensity. The results provide significant insights into how changes in oe production affect emission intensities.

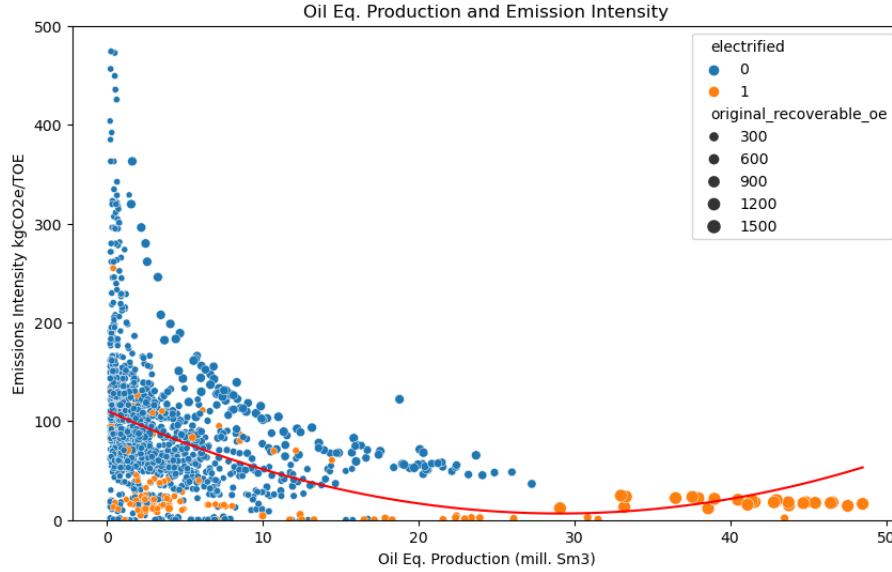


Figure 7: Oil Equivalent Production and Emission Intensity

7.1.3 DML 3 - Total Production on Emission Intensity

Key Findings The coefficient for *emission intensity* is -3.48, with a standard error of 1.12. This result is statistically significant, with a T-statistic of -3.11 and a P-value of 0.0019, indicating a strong negative relationship between production of hydrocarbons and emission intensity. The negative coefficient suggests that as oil equivalent production increases, the emission intensity decreases, reflecting improved operational efficiency at higher production levels.

The quadratic relationship for emission intensity is described by the equation:

$$y = 0.12x^2 - 7.17x + 110.79$$

	Emission Intensity	Emissions
Coefficient	-2.93	17987.67
Standard Error	0.844	3905.12
T-Statistic	-3.47	4.60
P-Value	0.0005	4×10^{-6}

Table 3: Summary of DML Results for Oil Equivalent Production and Emission Intensity

This fitted U-shaped curve indicates that emission intensity decreases with increasing production levels, reaching a minimum before marginally rising again as production continues to increase. The quadratic term is approximately ten times as high as the term for DML_1 and DML_2 . It is important to notice that most of the data points are concentrated on the left-hand side of the graph, where production volumes are low, and emission intensities are relatively high. Lower production levels tend to be less efficient in terms of emissions per unit of production. As production increases, the emission intensity decreases, with the lowest intensities observed in the middle of the production range (around 30 million Sm^3). However, the graph shows that there are fewer data points in this middle range, and all coming from electrified fields marked by the orange dots.

Interestingly, there seems to be a pattern between the fields’ original recoverable size and its emission intensity. In Figure 7, this is displayed by the size of the scatters. This relationship was also examined but not found in DML_1 and DML_2 . Figure 7 show that bigger fields (in terms of original recoverable size) follows a trend line that never drop below 60-70 $kgCO_2e/toe$ unless the field is electrified. Regarding new and future field on the NCS, many conceptions circle around developing large oil fields to keep emission intensities down. This study holds that true only if these large fields are electrified. Otherwise, smaller fields seem to be efficient. This is further backed up by the Causal Discovery and Figure 4, where there is no direct link between *Original recoverable oe* and any emission intensity parameters. Initially, one might think that it is due to smaller fields not processing their own products, but this is taken into account, as we use the distributed emission intensities as described earlier. Additional factors, as Sigrid Da Costa (Explorationist at Harbour Energy) suggested, may be related to the fact that new fields often consist of smaller reserves that were previously not economically viable, where newer technology and Machine Learning seismic discovery techniques has reduced previous uncertainties. However, these areas now benefit from existing production and extraction infrastructure, which facilitates their development.

7.2 Two-Way Fixed Effects

The results from the Two-Way Fixed Effects model provide a nuanced understanding of the impact of electrification on three different emission metrics related to oil and gas production. The analysis sheds light on differences between direct and distributed emission intensities and overall CO_2e emissions. The panel data consists of 79 different oil and gas fields, with 1093 observations over the 27 year period.

7.2.1 Direct Emission Intensity

Direct emission intensity refers to the emissions directly associated with the oil and gas extraction processes on a per-unit production basis within each field. Unlike distributed emission intensity,

which accounts for emissions spread across various operational aspects, including processing emissions from other facilities, direct emission intensity focuses solely on the emissions generated by the field's own activities. The direct emissions are often referred to as point source emissions and does not necessarily align with the "Polluter Pays Principle" (*European Court of Auditors*, 2021). This insight and distinction is crucial because many fields on the Norwegian Continental Shelf process and emit on behalf of other facilities, leading to differing impacts of electrification on these two metrics.

The results show a positive coefficient for direct emission intensity without controls (116.27), indicating an increase of 104% in emission intensity associated with electrification. When controls are introduced, this effect is somewhat mitigated, with the coefficient decreasing to 95.93, corresponding to an 85% increase. This increase in direct emission intensity, even after accounting for various field-specific factors, suggests that electrification might lead to higher emissions when we don't consider distributed emissions. Capturing the electrification effect solely on direct emissions would hence paint a disproportional image.

7.2.2 Emission Intensity (Distributed)

Distributed emission intensity takes into account emissions across the entire production network including hydrocarbons processed on behalf of other facilities, showed a substantial decrease following electrification. Without controls, the coefficient is negative at -47.30. This reduction becomes even more pronounced when controls are applied, with the coefficient dropping further to -51.51, with decreasing P-values. These coefficients translates to a 53% percent reduction in emission intensity compared to the non-treated mean. This is particularly interesting as it indicates that, when accounting for relevant controls, electrification has a strong potential to lower emissions intensity across various operational and field specific aspects. The p-values associated with these coefficients are highly significant (0.05 and 0.001 respectively), underscoring the robustness of these results.

7.2.3 Yearly CO_2e Emissions

Yearly CO_2e emissions show a substantial decrease associated with electrification. The coefficient for this variable is -111,424.10 without controls, corresponding to a 42% reduction in CO_2e emissions compared to the non-treated mean. When controls are introduced, the coefficient adjusts to -98,210.74, resulting in a 37% reduction. These insights highlight the effectiveness of electrification in lowering GHG emissions on an annual basis, though the magnitude of the reduction is slightly less pronounced when considering other influential factors. This consistent reduction across both models aligns well with policy goals aimed at minimizing the carbon footprint of oil and gas operations. Although it does not alter any Scope 3 emissions and it assumes that the electricity supplied to the fields is renewable. Furthermore, it assumes the absence of opportunity emissions, meaning no potential emissions are avoided by opting for a cleaner energy source. For example, using renewable electricity to produce green hydrogen could reduce the carbon footprint more than electrifying fossil fuel extraction using the same power.

7.2.4 Comparison and Implications

Overall, the TWFE model results illustrate the dual relationship between electrification and emission outcomes. The non-significant positive coefficient for emission intensity, even after controls are applied, suggest that while electrification may lead to some increases in certain types of emissions, its broader impact, particularly on distributed emission intensity and yearly CO_2e emissions, is decidedly negative. The significant reduction in these latter metrics, especially when controls are

accounted for, reinforces the potential of electrification as a viable strategy for decarbonizing oil and gas operations.

However, the variance in impact across different emission metrics indicates that electrification can have unintended unfair consequences, as it would benefit operators differently NCS depending on their direct and distributed emissions.

In terms of percentage change, emission intensity and annual absolute emissions diverge when production controls are taken into account. Specifically, while the reduction in emission intensity becomes more pronounced, the reduction in absolute emissions decreases. This difference highlights that even when controlling for production, they may achieve lower emission intensities because the fixed emissions (e.g., from infrastructure) are spread over a greater output. However, total annual emissions may still increase with higher production volumes, even if emission intensity per unit decreases. Controlling for production isolates the impact of variable emissions, potentially diminishing the perceived reduction in total emissions by emphasizing the relationship between production scale and absolute emissions.

Dependent Variable	Coefficient	Percentage Change
Direct Emission Intensity	116.27	104%
	95.92 (with controls)	85%
Emission Intensity	-47.30**	-48%
	-51.51*** (with controls)	-53%
Annual Emissions	-111,424.10**	-42%
	-98,210.74** (with controls)	-37%
Model Statistics		
R-squared (Overall)	0.3999	
R-squared (Within)	0.2168	
Number of Observations	1,093	
Number of Entities	79	
F-statistic	9.4048 (p = 0.0000)	
Avg. Observations per Entity	13.835	
Time Periods	27	

Table 4: Summary of TWFE Model Results and Model Statistics. Coefficients for the models with controls are indicated in parentheses. Statistical significance is denoted as follows: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.001$.

7.3 Emissions and Production Predictions

The future of oil and gas production on the NCS is characterized by a rapid decline in production from current open fields as displayed in Figure 8. The projections show that a few fields like Troll, Johan Sverdrup, and Snøhvit has the highest cumulative production by a big margin. Troll has 624 million oil equivalents (oe) Sm^3 of cumulative production, followed by Johan Sverdrup's 272 million oe Sm^3 and Snøhvit's 152 million oe Sm^3 . These figures showcase the big scale of these fields, but they also highlight the concentration of production within a handful of sites.

However, the overall trend for these fields are clear: production is set to decline steeply when approaching 2030. By that time half of the current remaining reserves will be extracted. The

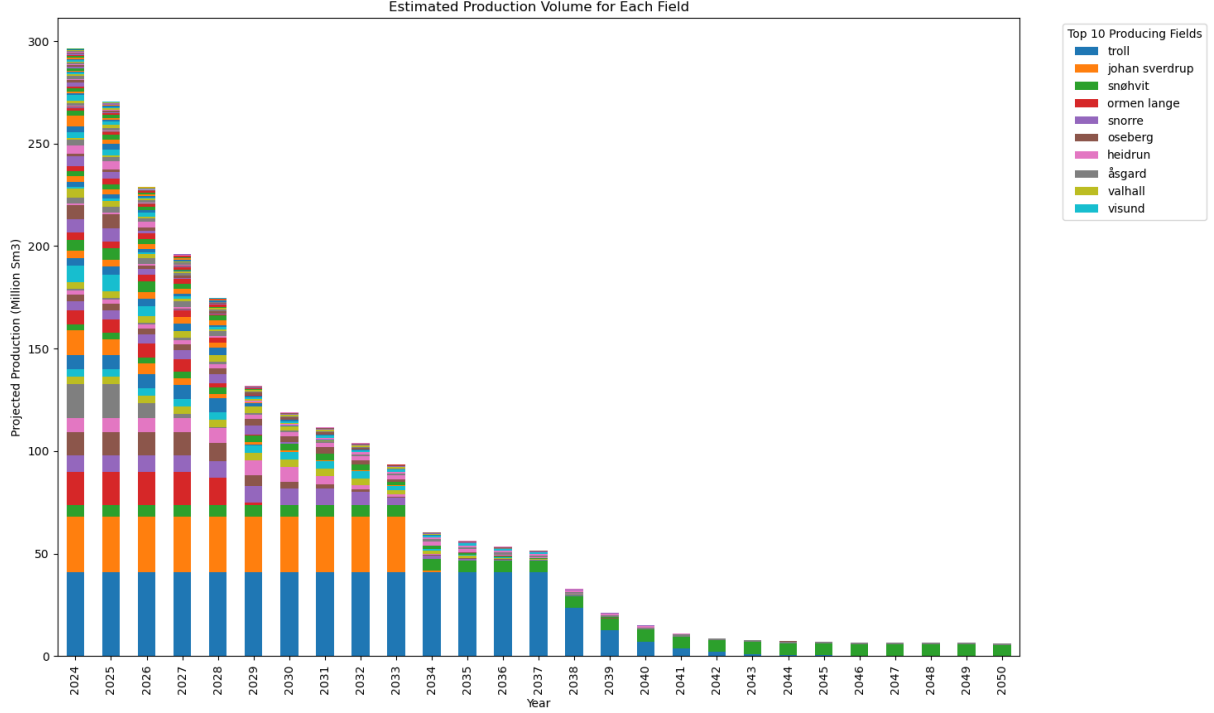


Figure 8: Estimated Oil Equivalent Production

average annual production across the fields stands at approximately 1.58 million oe Sm^3 , with significant variability—ranging from minimal outputs to as high as 40.93 million oe Sm^3 per year (Troll). This wide range suggests that while some fields will continue to be significant producers for the near term, others are rapidly approaching the end of their productive life.

On the emissions front fields such as Troll, Johan Sverdrup and Snøhvit are top producers; while Troll is also a top emitter, even it is already electrified. Highest in lifetime emissions is Ekofisk, responsible for over 132 thousand tons of CO_2e . These fields are significant contributors to the NCS’s overall carbon footprint.

We have showed that emissions are tightly coupled with production, hence, not unexpectedly, we see an uneven distribution in cumulative emissions as well. A few large fields are driving the majority of emissions, while others contribute relatively little. This uneven distribution points to an opportunity to target emission reductions in the highest-emitting fields. The next chapter will focus on optimizing production and emissions on the NCS.

Field	$oeMSm^3$
Troll	624
J. Sverdrup	273
Snøhvit	152
Ormen l.	79
Snorre	77

Table 5: Top Producers

Field	tCO_2e
Ekofisk	132,216
Oseberg	128,933
Åsgard	114,170
Troll	79,901
Gjøa Sør	67,594

Table 6: Top Emitters

Field	tCO_2e/toe
Snøhvit	1
Embla	3
Trym	3
J. Sverdrup	4
Ormen L.	4

Table 7: Lowest Intensities

7.4 Optimizing Production and Emissions

In the context of transitioning to a lower-carbon future, one critical challenge for NCS is balancing the dual objectives of maintaining high production levels while significantly reducing GHG emissions. To achieve this balance, an optimization model was employed, utilizing a range of lambda (λ) values to determine the optimal trade-off between production and emissions. The approach involves selectively shutting down the most polluting fields while prioritizing fields that offer higher production with lower emissions on the lifetime scale.

7.4.1 Reduction Scenarios

The optimization model produced several scenarios, each representing a different balance between production and emissions. The results indicate that as the lambda value increased, prioritizing production, fewer fields were shut down, resulting in higher overall production but also higher emissions. Conversely, lower lambda values, which prioritized emission reductions, led to more fields being shut down, significantly reducing emissions but also cutting production.

- **Scenario 1: Full Production Focus** ($\lambda = 1$) This scenario fully prioritized production with no emphasis on emission reductions. As a result, there was no reduction in production, maintaining a 0% production reduction and a 0% emission reduction. The estimated emission intensity was the highest at 39 kgCO₂e/toe. This is the baseline emission scenario, any measures of reductions in emissions are made by comparing to this full production focus.
- **Scenario 2: High Production Focus** ($\lambda = 0.9$) In this scenario, the focus remained heavily on production, leading to a minimal production reduction of just 1% and an emission reduction of 6%. The estimated emission intensity was 37 kgCO₂e/toe, indicating that while production was largely preserved, emissions were only slightly reduced.
- **Scenario 3: Balanced Approach** ($\lambda = 0.875$) With a more balanced focus between production and emissions, this scenario resulted in a production reduction of 7% and an emission reduction of 24%. The estimated emission intensity was 32 kgCO₂e/toe, reflecting a strategy that carefully weighs production needs against emission reduction efforts.
- **Scenario 4: Medium Emission Priority** ($\lambda = 0.85$) This scenario leaned slightly more towards emission reduction while still prioritizing production. The model suggested a production reduction of 15% alongside an emission reduction of 43%. The estimated emission intensity was 26 kgCO₂e/toe, demonstrating a moderate trade-off between production and environmental impact.
- **Scenario 5: Higher Emission Priority** ($\lambda = 0.8$) In this scenario, the model favored emission reduction more strongly. The result was a production reduction of 28%, with a significant emission reduction of 68%. The estimated emission intensity was 17 kgCO₂e/toe, showing that this approach effectively reduced emissions while still maintaining a moderate level of production.

The optimization results clearly demonstrate that significant reductions in emissions can be achieved on the NCS by strategically shutting down high-emission and low producing fields while maintaining production from more efficient, lower-emission fields. The balanced approach (Scenario 3) achieved a total of 24% emission reduction in lifetime emissions coming from the NCS compared to the full production focus, while maintaining a production at 93% of full capacity. By doubling the production reduction we are able to reduce emissions by 68% as described in the higher emission priority scenario (Scenario 5). With this pragmatic perspective on emission reductions, policymakers would be informed on what different phase-down scenarios would imply, and concrete numbers on

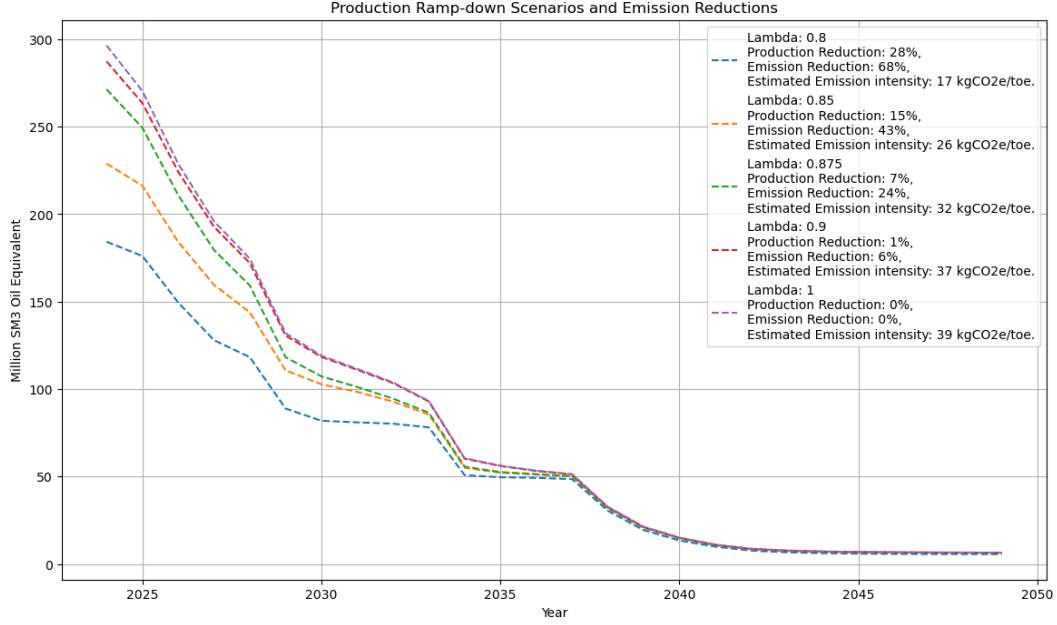


Figure 9: Estimated Oil Equivalent Production and Emission Reductions

emission reduction until 2050. Furthermore, the timing of the majority of emission reductions were investigated, and the larger share of emission cuts would happen before 2030, indicating a strong immediate effect. Figure 9 shows that all scenarios will eventually meet at 80 million Sm^3 of oil equivalents around 2033.

When plotting these fields geographically, Figure 10 shows that all fields for Scenario 2 are located in the North Sea with some proximity. Oil and gas fields in the Norwegian Sea are set to be closed when λ approaches 0.85. Again we find clusters of fields to be closed, suggestion a practical dimension to these theoretical results.

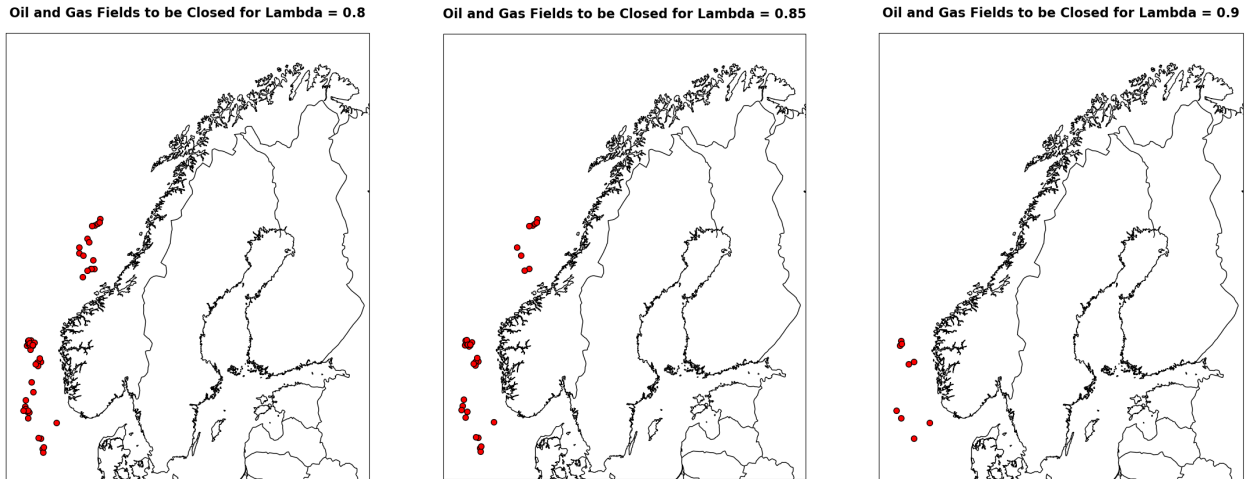


Figure 10: Closed O&G Fields for Scenario 5 ($\lambda = 0.8$), 4 ($\lambda = 0.85$) and 2 ($\lambda = 0.8$)

In the appendix follows an exhaustive list over decommissioned fields that correspond with the different scenarios and λ -values.

7.5 Electrification Optimization Results

Following the decommissioning analysis, this research explored the potential of electrification to reduce emissions while maintaining production. The optimization targeted fields for electrification across various levels; 0%, 10%, 15% and to 20%, translating to; 0, 3, 6, 10 and 13 fields.

At a 5% electrification level, targeting Heidrun, Oseberg and Visund yields a 13% reduction in emissions. This approach contrasts with the option of shutting down these fields entirely, which would eliminate their production output and potentially lead to economic losses. As the electrification level increases to 10%, additional fields like Ekofisk, Eldfisk and Njord are electrified, resulting in a 30% reduction in emissions. By expanding electrification to 15%, which includes major fields such as Gullfaks Sør, Oseberg Sør, Statfjord and Åsgard, emissions can be reduced by 42%. This level of reduction approaches what could be achieved through decommissioning of around 20 fields, coming with a production loss of 15% compared to full production. At the highest level of electrification, 20%, electrifying 13 fields, including Balder, Dvalin and Skarv, achieves a 51% reduction in emissions.

The graph in Figure 10 initially shows a steeper increase in emission reductions as the electrification level rises from 0% to 10%, followed by a more gradual slope as electrification levels continue to increase up to 20%. The steeper slope in the initial portion of the graph suggests that the early stages of electrification yield substantial emission reductions. This can be attributed to the fact that the first fields selected for electrification are likely the largest emitters or those with the highest emission intensities relative to their production. Electrifying these key fields results in significant decreases in total emissions, offering a strong return on investment for these initial efforts. As the electrification level increases beyond 20%, the graph begins to flatten, indicating a diminishing marginal return in terms of emission reductions. This level of electrification would require tremendous amount of electricity, and any further electrification can be questioned whether practically plausible. Even though, it is interesting to show that as more fields are electrified, the additional fields contribute progressively less to the overall emission reduction. This could be because the fields selected later in the process may have lower emission intensities, are already more efficient, or produce less, so electrifying them yields smaller gains in terms of emission reductions.

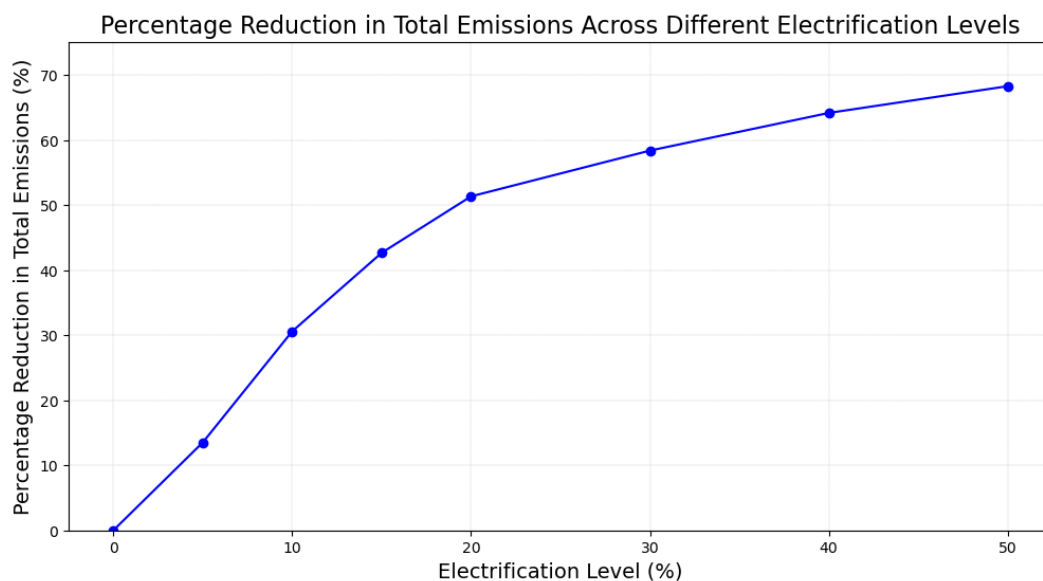


Figure 11: Estimated Oil Equivalent Production and Emission Reductions

8 Conclusion

The global energy transition puts pressure on oil and gas-producing nations to significantly reduce greenhouse gas emissions while maintaining energy security and economic stability. Norway faces the challenge of balancing its role as a major energy supplier with its commitment to cut emissions to meet reduction targets.

This thesis builds upon the existing literature, to explore how Norway can combine electrification and field decommissioning on the Norwegian Continental Shelf. This research is the first to use dual-objective numerical optimization in combination with Double Machine Learning to predict and calculate lifetime emissions for current open oil and gas fields. The results provide actionable insights into the optimization of emission reductions from Norway’s oil and gas sector with electrification and phase-out as means to balance production while reducing emissions.

The electrification-focused approach emerges as a promising pathway, especially in the context of Norway’s abundant renewable energy resources and the overall global focus on energy security. We find that direct point source emissions rise with field electrification, although not significant. But if the emissions are distributed according to production share and processing capabilities, a highly significant negative effect is found. By electrifying key fields such as Heidrun, Oseberg and Visund, 13% reduction in emissions can be achieved with minimal disruption to production. Extending this strategy to include additional fields like Ekofisk, Eldfisk, Gullfaks Sør, Statfjord and 6 others can push this reduction to 51%. Simultaneously, the power and grid requirements for such a widespread electrification are huge, exemplified by one of the largest electrified fields Troll which pulls around 1300 GWh annually.

In contrast, the decommissioning focused approach presents a more pragmatic strategy for reducing emissions. This approach comes with a trade-off in terms of production. For instance, a scenario that heavily prioritizes emission reduction could see a 68% cut in emissions by shutting down 49 fields and reducing production by 20% compared to the business as usual case, where all wells and fields are pumped dry. This finding is consistent with the broader research consensus, which acknowledges that decommissioning is one of the most effective ways to reduce emissions (*International Institute for Sustainable Development*, 2024).

On the flip-side, decommissioning poses economic and supply-demand challenges. Rystad Energy highlighted in 2021 that global emissions would rise if Norway ceased their oil and gas extraction and production (*Rystad Energy*, 2021). This occurs due to production shifting to other emission-intensive oil and gas producers. On the other side, Vista Analyse on behalf of the World Wildlife Fund and Greenpeace found the opposite: increase in Norwegian oil and gas production pushes global fossil fuel consumption (*Vista Analyse*, 2023). Hence a balanced approach is needed, and we found that a 15% reduction in production would yield a 43% reduction terrestrial oil and gas operational emissions, measuring up to the same emission cut from electrifying 10 fields.

This thesis describes how emission reduction efforts would require decommissioning fields first in the North Sea and later in the Norwegian Sea. A predictive scenario suggests a natural near-complete Norwegian Continental Shelf decarbonization by 2048, if no new wells are drilled. If new fields are to be developed, our prescription is to focus on smaller fields with existing infrastructure, ensuring that future projects are aligned with stringent emission reduction targets to limit global warming.

9 Discussion and Limitations

9.1 Combining Electrification and Decommissioning

One of the central conclusions of this study is that electrification, when applied selectively, can achieve significant emission reductions without the immediate economic drawbacks associated with decommissioning. However, decommissioning remains a critical tool for achieving deeper emissions cuts, especially for fields that are less efficient or nearing the end of their productive life.

The discussion should advocate for a hybrid strategy that integrates selective electrification with strategic decommissioning. By electrifying fields that are still economically viable and strategically important, Norway can continue to benefit from oil and gas revenues while significantly lowering its carbon footprint. Concurrently, decommissioning fields that are either high emitters or low producers could complement electrification efforts, resulting in a more comprehensive reduction in emissions.

9.2 Economic and Global Market Considerations

While the environmental benefits of combining electrification and decommissioning are clear, the economic implications must also be carefully considered. The oil and gas industry is a cornerstone of the Norwegian economy, and any significant reduction in production could have consequences. However, the shift towards more sustainable practices could also present new opportunities, like maintaining production while reducing terrestrial Scope 1 and Scope 2 emissions.

9.3 Opportunity Emissions

The concept of "opportunity emissions" expands the traditional focus on direct and distributed emissions by considering the emissions that the used renewable energy could otherwise be used to. As technologies such as green hydrogen develop, this dimension will be able to measure to provide a more nuanced view on how best to use the renewable power at disposal. In this thesis, the primary focus was on reducing direct and distributed emissions through electrification and optimization of production strategies. While the concept of opportunity emissions was not directly applied, incorporating it into future analyses could provide a more comprehensive understanding of the environmental impacts and help align industry practices with long-term sustainability goals.

9.4 Consideration beyond Carbon Dioxide and Methane

This research focuses on carbon dioxide and methane emissions, which are the most significant contributors to GHG emissions from this sector. However, nitrous oxide emissions have not been included in the analysis due to the lack of available data. These emissions, though typically smaller in volume, have a much higher global warming potential and could contribute significantly to the overall environmental impact.

Additionally, this study does not address other critical environmental factors beyond GHG emissions. For instance, oil spills, which are documented by authorities such as Norskeutslipp, represent a major environmental hazard, particularly to ocean and coastal ecosystems. Future research could expand on this work by incorporating data on oil spills and other pollutants to evaluate the broader environmental health impacts of oil and gas activities. Such an approach would provide a more holistic view of the sustainability challenges associated with hydrocarbon extraction and could guide more comprehensive environmental policy and mitigation strategies.

10 Appendix

10.1 Production and Emission Optimization

λ	Prod. Cut	Em. Cut	# Closed	Closed Fields
0.80	28%	68%	49	Alve, Balder, Bauge, Brage, Draugen, Dvalin, Ekofisk, Eldfisk, Fenja, Gimle, Gina Krog, Gudrun, Gullfaks, Gullfaks Sør, Gungne, Heidrun, Hyme, Kristin, Marulk, Mikkell, Morvin, Njord, Norne, Oda, Oseberg, Oseberg Sør, Oseberg Øst, Rev, Sigyn, Skuld, Sleipner Vest, Sleipner Øst, Snorre, Statfjord, Statfjord Nord, Statfjord Øst, Sygna, Tor, Tordis, Trestakk, Tune, Ula, Urd, Utgard, Vigdis, Vilje, Visund, Visund Sør, Yme
0.85	15%	43%	34	Alve, Brage, Ekofisk, Eldfisk, Gimle, Gina Krog, Gudrun, Gullfaks, Hyme, Marulk, Morvin, Njord, Norne, Oda, Oseberg, Oseberg Sør, Oseberg Øst, Rev, Sigyn, Skuld, Statfjord, Statfjord Nord, Statfjord Øst, Sygna, Tor, Tordis, Trestakk, Tune, Ula, Urd, Utgard, Visund, Visund Sør, Yme
0.875	7%	24%	20	Alve, Brage, Eldfisk, Marulk, Njord, Oda, Oseberg Sør, Oseberg Øst, Rev, Skuld, Statfjord, Statfjord Nord, Statfjord Øst, Sygna, Tor, Tune, Ula, Utgard, Visund, Yme
0.90	1%	6%	9	Brage, Oda, Rev, Statfjord, Statfjord Nord, Statfjord Øst, Tune, Utgard, Yme
1.00	0%	0%	0	None

Table 8: Fields Closed at Different Optimization Scenarios

10.2 Electrification Optimization ($\lambda = 0.8$)

Electrification Level	Emission Cut	Fields Electrified
5% (N=3)	13%	Heidrun, Oseberg, Visund
10% (N=6)	31%	Ekofisk, Eldfisk, Heidrun, Oseberg, Visund
15% (N=10)	43%	Ekofisk, Eldfisk, Gullfaks Sør, Heidrun, Njord, Oseberg, Oseberg Sør, Visund, Åsgard
20% (N=13)	51%	Balder, Dvalin, Ekofisk, Eldfisk, Gullfaks Sør, Heidrun, Njord, Oseberg, Oseberg Sør, Skarv, Statfjord, Visund, Åsgard

Table 9: Electrified Fields at Different Electrification Levels

10.3 Code

All the data, code, and methods are open for reuse and accessible through GitHub https://github.com/percw/Norwegian_oil_gas_decarbonization.

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